HEPro Business Analytics Project-2

HR Analytics Dashboard Development – Project Report

Fellow Name: Digvijay Singh
Program Name: HEPro Business Analytics Fellowship
Organization/Case: ZenithTech Solutions
Tool Used: Power BI
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2. Introduction

Human Resource (HR) Analytics is the systematic application of data analysis techniques to workforce-related information in order to improve organizational decision-making. Modern organizations generate vast amounts of employee data, including demographics, performance indicators, salary structures, tenure records, and attrition history. When properly analyzed, this data provides deep insights into workforce stability, employee engagement, and productivity trends.

This project focuses on the development of an HR Analytics Dashboard for **ZenithTech Solutions**. The primary aim is to transform raw employee data into structured, interactive visual insights using Power BI. The dashboard enables leadership to monitor attrition trends, analyze salary segmentation, evaluate demographic patterns, and identify early-warning indicators of workforce instability.

By leveraging analytics, the organization can move from reactive HR practices to proactive talent management. The dashboard serves as a strategic tool to reduce turnover costs, enhance employee satisfaction, and strengthen long-term workforce planning.

3. Project Objectives

The key objectives of this project are outlined below:

- Develop a comprehensive HR Analytics Dashboard using Power BI.
- Analyze 1,470 employee records to extract actionable insights.
- Identify patterns and drivers of employee attrition.
- Evaluate the impact of salary segmentation on retention.
- Examine demographic trends across age, gender, and education fields.
- Provide strategic HR recommendations based on analytical findings.

The dashboard is designed to support executive-level monitoring while also enabling detailed analysis through interactive filtering and cross-visualization capabilities.

4. Company Overview

ZenithTech Solutions is a mid-sized technology organization operating in a competitive digital services environment. The company employs a total workforce of **1,470 employees**, distributed across multiple departments including Sales, Research & Development, Human Resources, and Technical Operations. The workforce consists of professionals in key roles such as Sales Executives, Research Scientists, Developers, System Analysts, HR Managers, and Laboratory Technicians.

The organization currently records a **total attrition of 342 employees**, resulting in an overall **attrition rate of 23.27%**. This indicates that nearly one out of every four employees have exited the organization during the analysed period. Such a turnover level represents a significant operational and financial challenge, especially in high-demand technical roles.

From a demographic standpoint:

- The **average employee age is 38.81 years**, reflecting a predominantly mid-career workforce.
- The **average tenure is 6.96 years**, suggesting moderate employee retention but with notable early-career exits.
- The **31–45 age group shows the highest attrition levels**, indicating potential mid-career stagnation or competitive job market movement.
- Male employees account for approximately **61% of total attrition (209 exits)**, while female employees account for **39% (133 exits)**.

Compensation distribution reveals that employees in the **0–5K salary range contribute the highest attrition volume (188 exits)**, highlighting possible compensation dissatisfaction at entry and junior levels.

Role-based analysis shows:

- **Sales Executives (68 exits)** and
- **Research Scientists (63 exits)**

as the highest-risk job roles within the organization.

Given these workforce dynamics, ZenithTech Solutions faces key HR challenges including:

- Early tenure attrition
- Compensation competitiveness
- Mid-career retention gaps
- Role-specific workforce instability

To address these challenges, the organization has adopted a data-driven HR analytics framework. By leveraging Power BI dashboards and structured KPI monitoring, ZenithTech aims to reduce attrition, improve workforce engagement, and align HR strategy with long-term business objectives.

5. Dataset Description

The dataset used for this project is a structured HR dataset containing **1,470 employee records**, representing the complete workforce of ZenithTech Solutions during the analysis period. The dataset is simulated for analytical learning purposes but reflects realistic workforce patterns commonly observed in technology organizations.

The dataset serves as the foundation for building key HR metrics, performing workforce segmentation, and identifying attrition trends.

5.1 Data Source

- Source: Simulated HR Analytics Dataset

- File Format: Excel (.xlsx)
- Tool Used for Import: Power BI Desktop
- Total Records: **1,470 employees**
- Total Attrition Cases: **342 employees**
- Attrition Rate: **23.27%**

The dataset was imported into Power BI and renamed as **HR_Analytics** for modeling consistency.

5.2 Data Structure and Variables

The dataset consists of multiple categorical and numerical variables. These variables can be grouped into the following categories:

1. Employee Identification Fields

- **EmployeeID** – Unique identifier for each employee.
- Ensures accurate counting and prevents duplication.

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2. Demographic Variables

These variables help analyze workforce composition and diversity.

- **Age** (Numeric)
- **Gender** (Male/Female)
- **EducationField** (Life Sciences, Medical, Marketing, Technical Degree, etc.)

These variables enable demographic segmentation such as:

- Age-based attrition
- Gender-based turnover comparison
- Education field impact on attrition

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3. Job & Organizational Variables

These fields define the employee's role within the organization.

- **Department** (Sales, R&D, HR, etc.)
- **JobRole** (Sales Executive, Research Scientist, Developer, System Analyst, etc.)
- **YearsAtCompany** (Tenure in years)

These variables are critical for:

- Role-based attrition analysis
- Department-level workforce stability
- Tenure-based risk identification

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4. Compensation Variable

- **MonthlyIncome** (Numeric)

This variable was used to:

- Create salary segmentation
- Analyze compensation-driven attrition patterns
- Compare attrition across income levels

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5. Attrition Variable

- **Attrition** (Yes/No)

This is the **target variable** of the analysis.
All attrition-related KPIs and insights are derived from this field.

5.3 Derived Columns Created

To enhance analytical depth, calculated columns were created using DAX:

1. Age Group

Employees were segmented into:

- 16–30 (Early Career)
- 31–45 (Mid Career)
- 46+ (Senior Workforce)

Purpose:

- Identify high-risk age segments
- Study generational turnover patterns

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2. Salary Range

Monthly income was grouped into:

- 0–5K
- 5–10K
- 10K+

Purpose:

- Analyze compensation-related attrition
- Support salary benchmarking insights

These derived columns transformed continuous numerical data into categorical segments for easier visual comparison.

5.4 Key Measures Created (DAX Calculations)

To enable KPI-based dashboard monitoring, the following measures were created:

- **Total Employees** = COUNT(EmployeeID) → 1,470
- **Total Attrition** = 342
- **Attrition Rate** = 23.27%
- **Average Age** = 38.81 years
- **Average Salary** ≈ 11K
- **Average Years at Company** = 6.96 years

These measures dynamically update based on slicers and filters applied within the dashboard.

6. Data Preparation

Data Cleaning:

- Verified data types for numeric and categorical fields.
- Ensured consistency in attrition labels (Yes/No).
- Validated absence of missing or null EmployeeID values.

Data Transformation:

- Created Age Group classification using conditional logic.
- Created Salary Range categorization for compensation segmentation.

Calculated Fields (DAX Measures):

- Total Employees = COUNT(EmployeeID)
- Total Attrition = Count where Attrition = "Yes"

- Attrition Rate = Total Attrition / Total Employees
- Average Age
- Average Salary
- Average Years at Company

These measures enabled dynamic, real-time KPI updates within the dashboard.

7. Tools Used

Power BI

Power BI was used for:

- Data modeling
- DAX measure creation
- Interactive dashboard development
- Data visualization and cross-filtering

Microsoft Excel

Excel was used as the source dataset platform and for preliminary structure validation.

8. Dashboard Design

The dashboard was designed following a structured layout approach.

KPI Section (Top Panel)

- Total Employees: 1,470
- Total Attrition: 342
- Attrition Rate: 23.27%
- Average Age: 38.81 years
- Average Salary: 11.00K
- Average Years at Company: 6.96 years

Charts Used:

- Bar Chart: Attrition by Job Role
- Bar Chart: Attrition by Age Group
- Bar Chart: Attrition by Salary Range
- Donut Chart: Attrition by Education Field
- Pie Chart: Attrition by Gender
- Line Chart: Attrition Trend by Years at Company

Filters Applied:

- Department slicer (vertical orientation)
- Interactive cross-filtering between visuals

The design ensures clarity, consistency, and executive-level readability.

9. Analysis & Insights

Role-Based Attrition

Sales Executives (68) and Research Scientists (63) exhibit the highest attrition. These roles may face high external demand or performance pressure.

Age & Gender Analysis

The 31–45 age group shows the highest attrition (127). Male employees account for 61% of total attrition, indicating demographic concentration of exits.

Salary Analysis

The 0–5K salary range contributes the highest attrition (188). Lower compensation bands appear more vulnerable to turnover.

Tenure Analysis

Attrition trends indicate higher exits among employees with shorter tenure (early career risk). This suggests onboarding and engagement challenges within the first two years.

10. Key Findings

The HR analytics dashboard reveals several important workforce trends at ZenithTech Solutions based on the analysis of 1,470 employees and 342 attrition cases (23.27% attrition rate).

First, the overall attrition rate of 23.27% indicates a relatively high turnover level, suggesting workforce instability that may impact productivity, recruitment costs, and organizational continuity.

Role-based analysis shows that **Sales Executives (68 exits)** and **Research Scientists (63 exits)** experience the highest attrition. These roles are likely exposed to high performance pressure and strong external market demand, making them more vulnerable to turnover.

Salary segmentation highlights that employees in the **0–5K income range account for the highest attrition (188 exits)**. This suggests compensation dissatisfaction among entry-level or junior employees, indicating a need to reassess pay competitiveness.

Demographic analysis reveals that the **31–45 age group records the highest attrition (127 exits)**, pointing toward mid-career mobility or limited advancement opportunities. Additionally, male employees represent approximately **61% of total exits (209 cases)**, showing a demographic concentration in attrition patterns.

Tenure analysis indicates higher exit frequency within the **first two years** of employment, suggesting potential onboarding gaps or unmet expectations during early employment stages.

Overall, attrition at ZenithTech Solutions appears to be influenced by multiple factors including compensation levels, job role demands, career progression opportunities, and early tenure engagement. These findings highlight the need for targeted, data-driven retention strategies rather than uniform HR interventions.

11. Recommendations

Based on the analysis:

Retention Actions:

- Conduct stays interviews in high-risk roles.

- Implement structured onboarding for first 90 days.
- Establish mentorship programs for early-tenure employees.

Engagement Strategies:

- Provide mid-career growth pathways.
- Offer leadership development programs.
- Improve work-life balance initiatives.

Compensation Suggestions:

- Review salary structure for 0–5K bracket.
- Introduce performance-based incentives.
- Conduct salary benchmarking against industry standards.

These targeted interventions can significantly reduce attrition while improving workforce engagement.

12. Conclusions

Project Outcome

The HR Analytics Dashboard has successfully transitioned ZenithTech Solutions from an intuitive management style to a data-driven strategic model. By auditing 1,470 employee records and centralizing them into a Power BI environment, we have transformed raw data into a live "Organizational Health Monitor." The project successfully pinpointed that attrition is not a general problem, but one localized within specific salary brackets and early-tenure windows.

Business Value & Impact

The primary value of this dashboard lies in **Precision Intervention**:

- **Cost Savings:** By identifying the high turnover in the 0–5k salary range (188 exits), the company can now apply targeted financial adjustments rather than expensive, blanket pay raises.
- **Operational Continuity:** Recognizing the "early-tenure risk" for employees with <2 years allows HR to fix onboarding gaps, preserving the high cost of initial recruitment.
- **Strategic Foresight:** Leadership can now anticipate departures in high-pressure roles like Sales and Research, moving from reacting to resignations to preventing them through "Stay Interviews."

Professional Insights

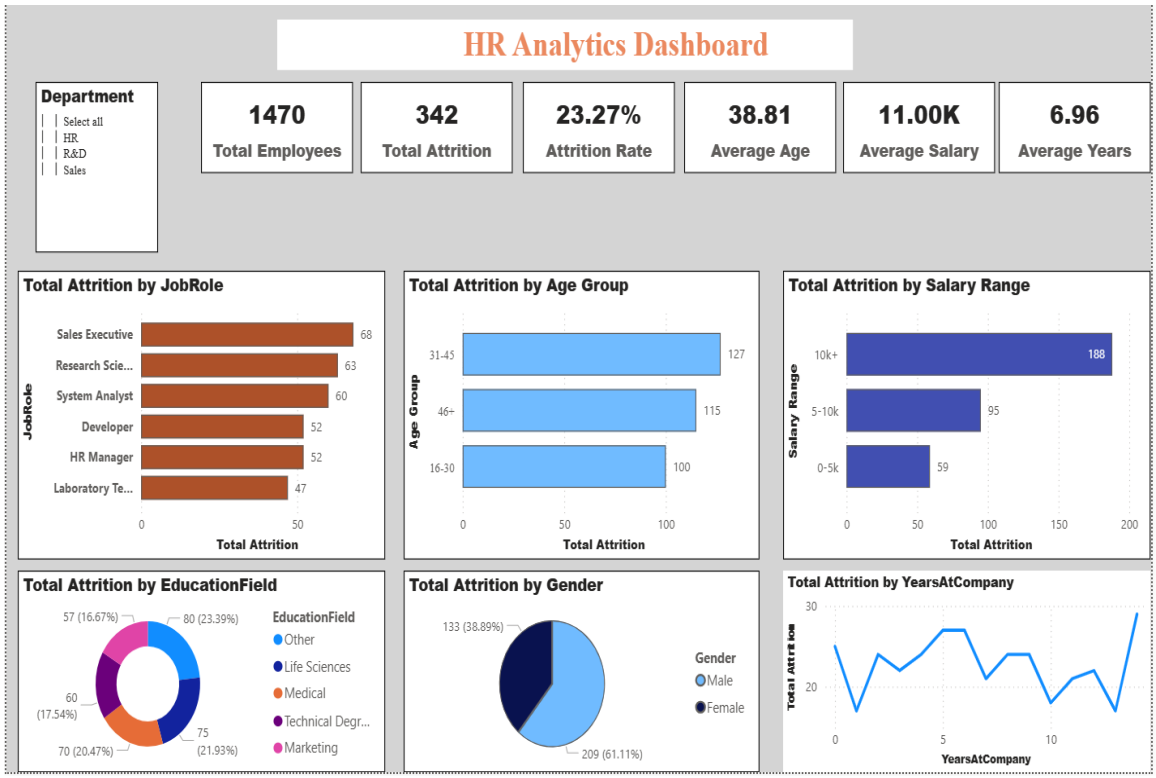
Through this project, the most critical learning was the power of **Salary Segmentation**. It revealed that financial dissatisfaction is a primary driver of instability at ZenithTech. Mastering the technical workflow—from data cleaning in Excel to creating complex DAX measures in Power BI—has provided the skills necessary to turn workforce statistics into a roadmap for organizational stability.

Final Statement

This project concludes that while ZenithTech faces a high attrition rate of 23.27%, the solution is reachable through targeted, data-backed policies. This dashboard provides the transparency needed to foster a more stable, satisfied, and cost-efficient workforce.

13. Appendix

Dashboard Screenshot



Dataset Structure Screenshot

EmployeeID	Age	Gender	Department	JobRole	EducationField	MonthlyIncome	YearsAtCompany	Attrition	Age Group	Salary Range
1008	40	Male	Sales	Sales Executive	Medical	4983	9	No	31-45	0-5k
1029	45	Male	Sales	Sales Executive	Medical	2834	2	No	31-45	0-5k
1044	19	Male	Sales	Developer	Medical	10526	7	No	16-30	10k+
1071	57	Male	Sales	Laboratory Technician	Medical	19095	9	No	46+	10k+
1086	18	Male	Sales	Developer	Medical	3777	1	No	16-30	0-5k
1091	25	Male	Sales	System Analyst	Medical	11496	0	No	16-30	10k+
1095	22	Male	Sales	Developer	Medical	10343	8	No	16-30	10k+
1102	25	Male	Sales	Developer	Medical	6133	7	No	16-30	5-10k
1139	32	Male	Sales	Sales Executive	Medical	17235	8	No	31-45	10k+
1143	49	Male	Sales	Sales Executive	Medical	13042	0	No	46+	10k+
1154	28	Male	Sales	Developer	Medical	2191	5	No	16-30	0-5k
1182	54	Male	Sales	HR Manager	Medical	19740	12	No	46+	10k+
1208	33	Male	Sales	System Analyst	Medical	17563	4	No	31-45	10k+
1214	20	Male	Sales	Laboratory Technician	Medical	8163	8	No	16-30	5-10k
1246	54	Male	Sales	Laboratory Technician	Medical	16330	8	No	46+	10k+
1272	52	Male	Sales	Developer	Medical	2889	0	No	46+	0-5k
1295	44	Male	Sales	Developer	Medical	10458	1	No	31-45	10k+
1357	31	Male	Sales	Research Scientist	Medical	11638	12	No	31-45	10k+
1427	56	Male	Sales	Sales Executive	Medical	3102	10	No	46+	0-5k
1435	53	Male	Sales	Developer	Medical	10780	7	No	46+	10k+
1456	50	Male	Sales	HR Manager	Medical	8597	14	No	46+	5-10k
1460	18	Male	Sales	Laboratory Technician	Medical	12794	9	No	16-30	10k+
1473	25	Male	Sales	System Analyst	Medical	14231	6	No	16-30	10k+
1523	31	Male	Sales	System Analyst	Medical	4196	12	No	31-45	0-5k
1527	22	Male	Sales	Sales Executive	Medical	2864	2	No	16-30	0-5k
1532	53	Male	Sales	Research Scientist	Medical	5003	9	No	46+	5-10k
1559	46	Male	Sales	HR Manager	Medical	2055	11	No	46+	0-5k
1579	56	Male	Sales	HR Manager	Medical	18682	1	No	46+	10k+
1596	49	Male	Sales	Laboratory Technician	Medical	18622	11	No	46+	10k+
1599	38	Male	Sales	System Analyst	Medical	19770	12	No	31-45	10k+
1604	36	Male	Sales	Developer	Medical	3996	3	No	31-45	0-5k
1624	45	Male	Sales	Developer	Medical	4367	9	No	31-45	0-5k
1667	52	Male	Sales	HR Manager	Medical	15824	11	No	46+	10k+
1668	18	Male	Sales	System Analyst	Medical	15444	9	No	16-30	10k+
1671	45	Male	Sales	Sales Executive	Medical	2539	3	No	31-45	0-5k
1672	34	Male	Sales	Research Scientist	Medical	12974	14	No	31-45	10k+
1681	27	Male	Sales	Research Scientist	Medical	12338	2	No	16-30	10k+